Programming task 5: Magnetospheric B-field

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Task: Calculate the magnetospheric currents

The task is to Calculate the magnetospheric currents from magnetic-field in the noon-midnight-north-south plane (the X-Z plane in GSM-coordinates).
The file noon-midnight-B-fields.mat contains arrays for the X and Z-components of the Tsyganenko-model magnetic field, BX and BZ, as well as the dipole field-components, BXd and BZd, in nT, at the GSM X and Z-coordinates given in the arrays Xnm and Znm (here we use RE as units for convenience).
The noon-midnight-B-fields.mat file contents are:

```
  Name        Size                Bytes  Class     Attributes

  BX        801x801             5132808  double
  BXd       801x801             5132808  double
  BZ        801x801             5132808  double
  BZd       801x801             5132808  double
  Xnm       801x801             5132808  double
  Znm       801x801             5132808  double
```

When we have B-field values on a dense enough grid to calculate the curl of the magnetic field we can use Amperé-Maxwell's law to calculate the current-density:

$$\nabla \times \mathbf{B} = \mu_0 \left(\mathbf{J} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right)$$

On the time-scales of B-field-variations in the magnetosphere we can safely ignore the $\partial \mathbf{E}/\partial t$ term.
The solar wind parameters for this case were:

```
BY_IMF = 0;     % Y-component of IMF (nT)
BZ_IMF = -2.1;  % Z-component of IMF (nT)
P_SW = 3.16;    % solar wind dynamic pressure (nPa)
Dst = 1;        % Dst-index
PS = 0*pi/180;  % Magnetic dipole tilt angle
```

Under these conditions ($B_Y = 0$ and zero dipole tilt angle) the current across the noon-midnight-plane is purely in the Y-direction, since the Tsyganenko-magnetosphere is to first order mirror-symmetric in the dawn-dusk-direction.
To calculate the current what needs to be done is to calculate the y-component of $\nabla \times \mathbf{B}$:

```
mu_0 = 4*pi*1e-7;       % Permeability [Vs/Am]
R_E  = 6.378e6;         % Radius of earth [m]
Iy = curl_y(Xnm*R_E,Znm*R_E,BX,BZ)/mu_0;
```

For this you should use the standard definition of the curl-operation. Here it is strongly prefereable to remove the dipole-component from the B-fields, that component are in principle curl-free, but when samlped at a cartesian grid like here that is not guaranteed to be "numerically" true. After doing this you should display the current with something like *pcolormesh* and label the different currents. For plots of parameters such as currents, that are "balanced around zero" in the sense that when the currents close the currents across a plane adds to zero, it is often best to display the current with a "diverging colourmap" centred around zero. Also make plots of J_y together with the arrow-plot (in matplotlib there is a "quiver" function) of the magnetic field (with suitable spacing between the arrows). Also try to reproduce the field-line-plots to get a solid grasp of the coupling between the currents and the magnetic fields. For field-line-plotting you should try to use the "streamplot" function, that calculates the tangent-curves of the B-field numerically.

Final hints and advice : The dayside ring-current should be about 20 nA/m^2 in the negative y-direction. For calculating the derivatives of $\mathbf{B}$ that goes into the curl-calculation, I suggest that you use the numpy "gradient" function. Here it is also important to keep in mind that the order matlab and python/numpy stores arrays and calculates the gradients might not be identical. Therefore you might have to test (or figure out) which permutation of parameters in to and out from the gradient function are the ones you need.
When you make quiver-plots/arrow-plots you should not put arrows at every point in the grid, start with a coarse step in both x and z-direction, then you can refine it as suitable. Also remember to make the arrows unit-lengths, otherwise the arrows from the near-Earth region will be very much longer than the arrows further out.

Background/Introduction

The rest of this document is just an automatically generated document from a matlab-script where the B-fields were created, with some figures illustrating the magneti-field-structure of the magnetosphere. They are included here to give you some hints, leads and suggestions for how things should look and how you could present your results.
This template script illustrates the magnetic field in near-Earth space. Earth's dipole-field dominates near-Earth-space, but its field-strength decreases as $1/r^3$. Far outside Earth the interplanetary magnetic field (IMF) will dominate. At the boundary between these regions "surface"-currents (currents in very/comparatively thin layers) are induced by the interaction between the solar wind and Earth's magnetic field. Additional currents flow inside the magnetosphere. These currents generate additional magnetic fields, combined with Earth's dipole-field these field determines the shape of the magnetosphere.

For the general structure of the magnetosphere researchers have combined satellite observations over a long period of time for a wide range of geophysical conditions. The B-field observations have been combined into models for the magnetospheric field, the most widely used model is the Tsyganenko-model, which makes a good estimate of the expected magnetospheric structure.

First show the combination of the dipole-field from Earth and a constant interplanetary magnetic field in the negative Z-direction. (GSM-coordinates: the origin of the GSM-coordinates are at the centre of Earth, with the positive X-axis along the line towards the center of the Sun. The Y-axis is defined as the cross product of the GSM X-axis and the magnetic dipole axis; positive towards the dusk-side. The Z-axis is defined as the cross product of the X and Y-axes, completing a right-handed cartesian coordinate system. The magnetic dipole axis lies within the XZ plane at all times, thus the Y and Z-axes rotates around the X-axis through the day with the rotation of Earth, since the magnetic dipole axis is shifted from Earth's rotational axis.)

Coordinates of noon-midnight plane (X-Z-plane)

To illustrate the B-fields and currents we create a cartesian grid in the X-Z-plane. We make the grid at regular points, and for convenience we use R_E as the length-unit (this needs to be kept in mind).

```
[Xnm,Znm] = meshgrid(-65:0.1:15,-40:0.1:40);
```

Solar-wind parameter definition

The Tsyganenko-model is parameterized with the solar wind dynamic pressure, the IMF B_Y and B_Z and the Dst-index. The Dst-index (disturbance-storm-time-index) is a measure of the intensity of the ring-current (with a small con-

tribution from the near-Earth part of the cross-tail current). An increase of the ring-current will lead to a reduction in the north-ward B-field at low latitudes, so enhanced ring-currents are reflected in a negative Dst-value, positive deviations in the northward B–field can be caused by compression of the day-side magnetopause. The index is derived from a network of 4 near-equatorial geomagnetic observatories.

For this case we look at an arbitrary set of parameters:

```
BY_IMF = 0;     % Y-component of IMF (nT)
BZ_IMF = -2.1;  % Z-component of IMF (nT)
P_SW = 3.16;    % solar wind dynamic pressure (nPa)
Dst = 1;        % Dst-index
% [P_SW,BZ_IMF] = meshgrid([1 2]*pSW,[1 2]*IMF_Bz);

PS = 0*pi/180; % Magnetic dipole tilt angle at current time CMP +/-10 (equinoxes) and +33/+1
% PS is the angle of between the Z-axis (GSM) and the dipole-axis

if ~exist('Bfields_calculatemd') % this to only calculate the Tsyganenko-B-fields once
  Bfields_calculatemd = 0;
end
if ~Bfields_calculatemd

  % this script is modified from one where there were multiple
  % IMF-parameters and that script looped over all of them. Here ii1 and
  % ii2 are the number of BZ_IMF and P_SW components.
  ii1 = 1; % size(P_SW,1):-1:1
  ii2 = 1; % size(P_SW,2):-1:1
  PARMOD = [P_SW(ii1,ii2),Dst,BY_IMF,BZ_IMF(ii1,ii2)];
  disp(PARMOD)
```

T96 and dipole magnetic fields in the X-Z noon-midnight plane

```
  for i1 = size(Xnm,1):-1:1
    for i2 = size(Xnm,2):-1:1
      if Xnm(i1,i2)^2+Znm(i1,i2)^2 > 1
        % B-field components induced by Magnetospheric currents + B(IMF)
        [BX(i1,i2),BY(i1,i2),BZ(i1,i2)] = T96([],PARMOD,PS,Xnm(i1,i2),0,Znm(i1,i2));
        % Dipole B-field components
        [BXd(i1,i2),BYd(i1,i2),BZd(i1,i2)] = T96_DIPOLE(PS,Xnm(i1,i2),0,Znm(i1,i2));
      end
    end
  end
  Bfields_calculatemd = 1;

end
```

Vector-plots of B-fields of Dipole + B(IMF)

```
phi360 = linspace(0,2*pi,361); % just array with 361 elements in 1-degree steps
i_dx = 12;  % step in X-direction
i_dz = 12;  % step in Z-direction
for ii1 = 1
  for ii2 = 1
    figure('position',[2   606   813   368]) % Nicely sized figure
    colormap(turbo)                           % nice colourmap
    Btot = sqrt( (BXd).^2 + (BZ_IMF(ii1,ii2)+BZd).^2 ); % magnetic field-strength
    Btot(~isfinite(Btot(:))) = 1;  % Set B-fields that are not finite to one
    Btot(Btot(:)==0) = 1;          % Set B-fields-strength that are zero to one, to avoid di
    % quiver-plot that with colouring-capabilities. In python you have a
    % more capable "quiver" function (if I interpret the documentation
    % correctly) Unfortunately it only works for 3-D vector-fields and here
    % the quick-and-dirty solution was to just feed the function zeros for
    % the Z and B-z inputs, and use the log of the B-field-strength
    q = quiver3_cc(Xnm(3:i_dz:end,3:i_dx:end),...
                   Znm(3:i_dz:end,3:i_dx:end),...
                   0*Znm(3:i_dz:end,3:i_dx:end),...
                   (BXd(3:i_dz:end,3:i_dx:end))./Btot(3:i_dz:end,3:i_dx:end),...
                   (BZ_IMF(ii1,ii2)+BZd(3:i_dz:end,3:i_dx:end))./Btot(3:i_dz:end,3:i_dx:end
                   0*BY(3:i_dz:end,3:i_dx:end)./Btot(3:i_dz:end,3:i_dx:end),...
                   max(-0.5,log10(Btot(3:i_dz:end,3:i_dx:end))));
    set(q,'LineWidth',0.5)
    view(0,90)
    set(gca,'XDir','reverse','fontsize',12)
    axis([-30 30 -21 21 -0.1 0.1])
    axis equal
    caxis([-0.5 max(max(log10(Btot(3:i_dz:end,3:i_dx:end))))])
    cblh = colorbar_labeled('nT','log');
    set(cblh,'position',get(cblh,'position')+[-0.01 0 0 0])
    xlabel('X [Re]')
    ylabel('Z [Re]')
    hold on
    fill(cos(phi360),sin(phi360),'g')
    tstr = sprintf('Earth Dipole-B-field combined with IMF B_Z: %3.2f nT',...
                   BZ_IMF(ii1,ii2));
    title(tstr)

    drawnow
    hold on
    plot3(cos(phi360),sin(phi360),sin(phi360),'k')

    gcapos = get(gca,'position');
    cblp = get(cblh,'position');
```

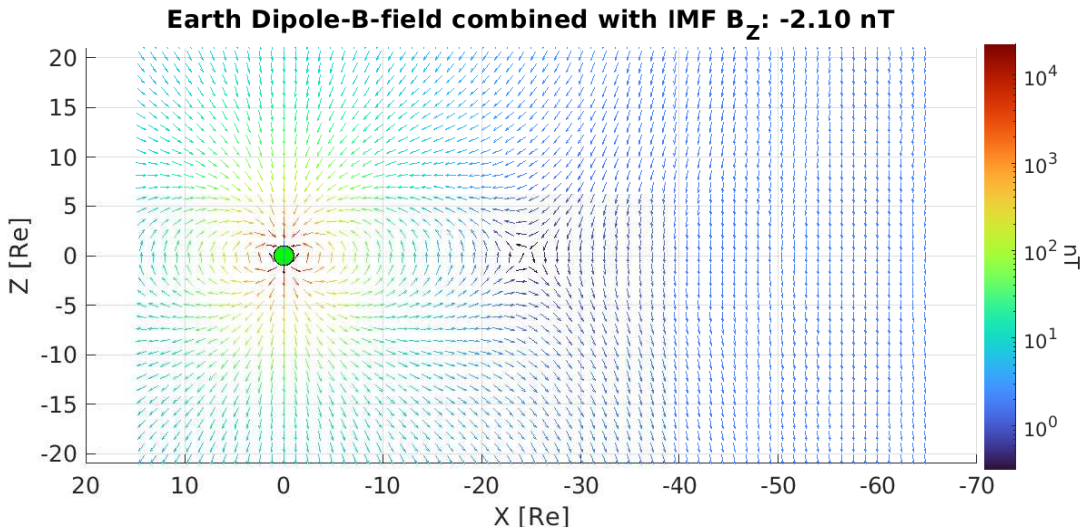

Figure 1: Note that the B-field at $Z = 0$ outsize of some 25 R_E are not much affected by Earth's dipole-field.

```
  end
end
```

Tsyganenko-model

```
figure('position',[2   321   805   653])
quiver3_c(Xnm(1:15:end,1:15:end),...
          Znm(1:15:end,1:15:end),...
          0*Znm(1:15:end,1:15:end),...
          (BX(1:15:end,1:15:end)+BXd(1:15:end,1:15:end))./ ...
          max(1e-4,(BZ(1:15:end,1:15:end)+BZd(1:15:end,1:15:end)).^2 + ...
                   (BX(1:15:end,1:15:end)+BXd(1:15:end,1:15:end)).^2).^.5,...
          (BZ(1:15:end,1:15:end)+BZd(1:15:end,1:15:end))./ ...
          max(1e-4,(BZ(1:15:end,1:15:end)+BZd(1:15:end,1:15:end)).^2 + ...
                   (BX(1:15:end,1:15:end)+BXd(1:15:end,1:15:end)).^2).^.5,...
          0*BZ(1:15:end,1:15:end),...
          log10(max(1e-0,(BZ(1:15:end,1:15:end).^2+BX(1:15:end,1:15:end).^2).^.5)),...
          turbo);
view(0,90)
set(gca,'XDir','reverse','fontsize',12)
hold on
plot(cos(phi360),sin(phi360),'k')
phE = fill(cos(phi360),sin(phi360),'b');
set(phE,'facecolor',[0.2 0.4 1])
fill(cos(phi360([91:270])),sin(phi360([91:270])),'k')
tstr = sprintf('Magnetospheric B-field (T-96 model)');
```

```
title(tstr)
xlabel('X [Re]')
ylabel('Z [Re]')
```

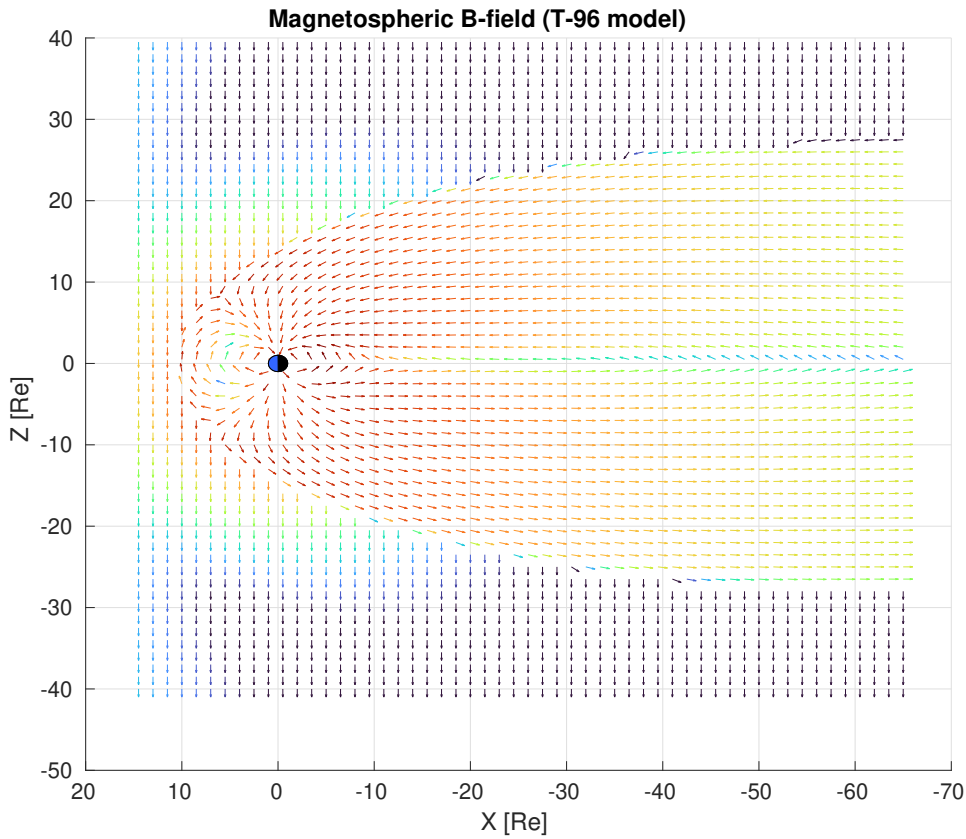

Figure 2: The Magnetospheric B-field looks more like this, in the Tsyganenko-model

"Open" Field-lines from IMF into Magnetosphere from North

In the lectures we talked about how to calculate the field-lines for the B-field from a magnetic dipole, using some nice geometry it was possible to obtain a a "nice and simple" ODE for the radius of a field-line as a function of magnetic latitude, that lead to a nice and simple solution for the magnetic field. The Tsyganenko-model have functions to solve the much gnarlier ODEs for the field-lines with the contributions from the IMF and the magnetospheric currents.

Here we instead use a much simpler way to calculate the field-lines using the "streamline" function (matlab, in python/matplotlib it seems that there is a

much more potent function: streamplot)

```
xSN = [-62:10:2,5 8 9 10 11 12];
zSN = 40*ones(size(xSN));
hS = streamline(Xnm,Znm,BX+BXd,BZ+BZd,xSN,zSN);
set(hS,'color','k')
ax0 = axis;
```

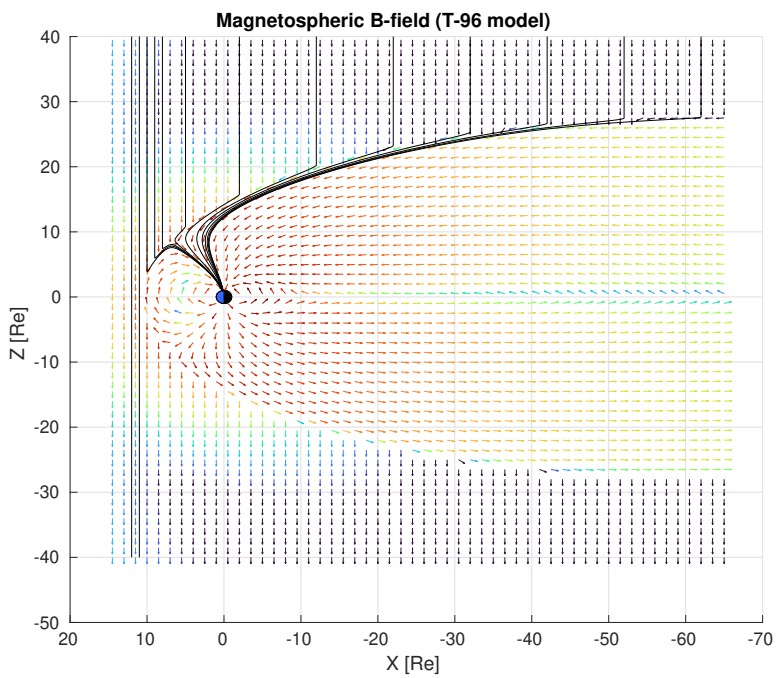

Figure 3: Note how the magnetic field-lines (tangent-curves of the magnetic field) from interplanetary space are funneled into a very small region on the day-side edge of the polar cap at ionospheric heights. This general region are known as the "cusp"-region, magnetospheric and ionospheric cusp, respectively.

```
axis([-5 12 0 10])
```

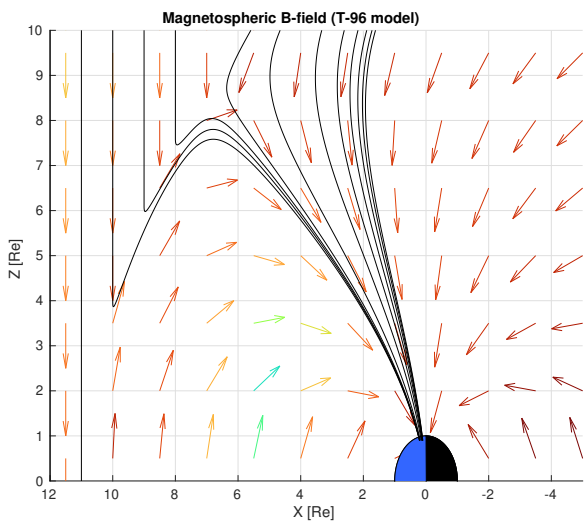

Figure 4: The ionospheric cusp-region have foot-points of magnetic field-lines that extend out into the solar wind - this means that plasma can move along the field-lines directly from the solar wind down into the ionosphere in this region on the day-side.

Polar-cap/magnetotail-lobe field-lines

```
zST = [1:26];
xST = -65*ones(size(zST));
hS = streamline(Xnm,Znm,BX+BXd,BZ+BZd,xST,zST);
set(hS,'color',[0.4 1 0.4])
```

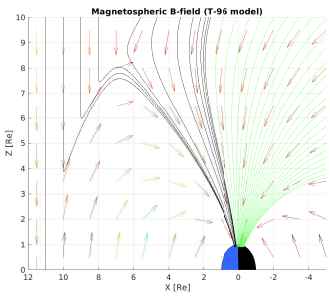

Figure 5: While the magnetic field-lines with foot-points in the polar cap...

```
axis(ax0)
```

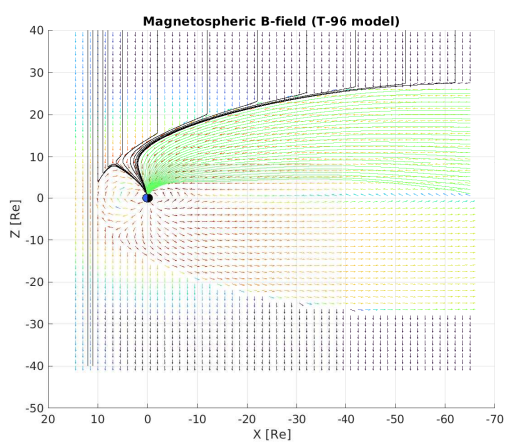

Figure 6: ...extends out into the northern (and southern) lobes of the magnetotail

Plasma-sheet field-lines

```
xSP = -60:2:-8;
zSP = zeros(size(xSP));
hS = streamline(Xnm,Znm,BX+BXd,BZ+BZd,xSP,zSP);
```

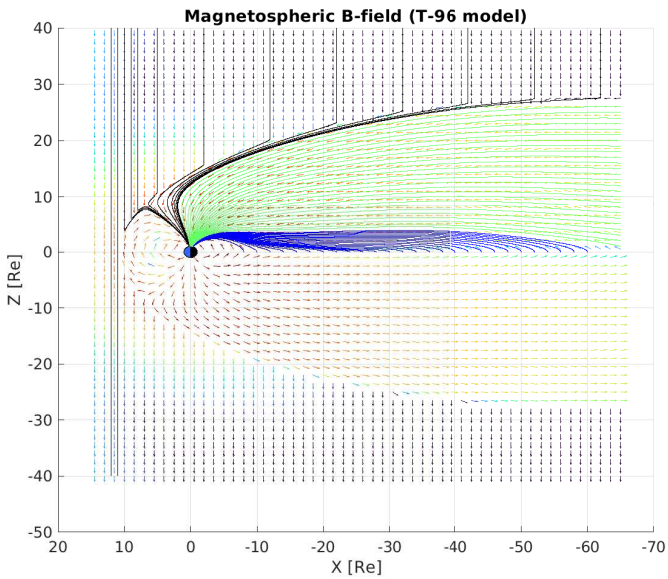

Figure 7: Field-lines from the equatorial region of the magneto-tail maps down into the night-side edge of the polar cap. This corresponds to the auroral oval, at least the night-side part of the auroral oval.

Closed field-lines and Southern polar cap-region

```
xSE = 2*cos(phi360([10:10:70,73:2:105,130:10:170]));
zSE = -2*sin(phi360([10:10:70,73:2:105,130:10:170]));
hS = streamline(Xnm,Znm,BX+BXd,BZ+BZd,xSE,zSE);
set(hS,'color','m')
```

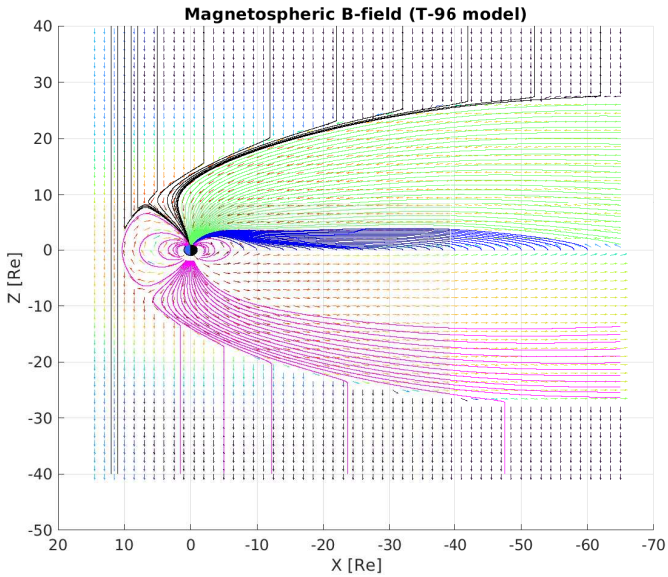

Figure 8: From lower magnetic latitudes we have field-lines that do not extend too far from Earth and have roughly dipolar shape (though slightly blunted and squished), and from the southern polar cap we have field-lines that extend out into the solar wind and the southern magneto-tail.